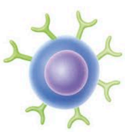
White blood cells

White blood cells are like warriors fighting viruses, bacteria, and cancer cells in your body. Fragments of the invaders, called antigens, are presented so the warriors can strategically fight using the right antibodies and chemical messengers, called cytokines. Communication is key—miscommunication can lead to chronic inflammation.

DENDRITIC CELL
These present antigens, which the body recognizes as a foreign invader. They activate T-cells to do their job.

MACROPHAGE
Hungry hunter cells (see phagocytosis below) that also release cytokines to induce inflammation.



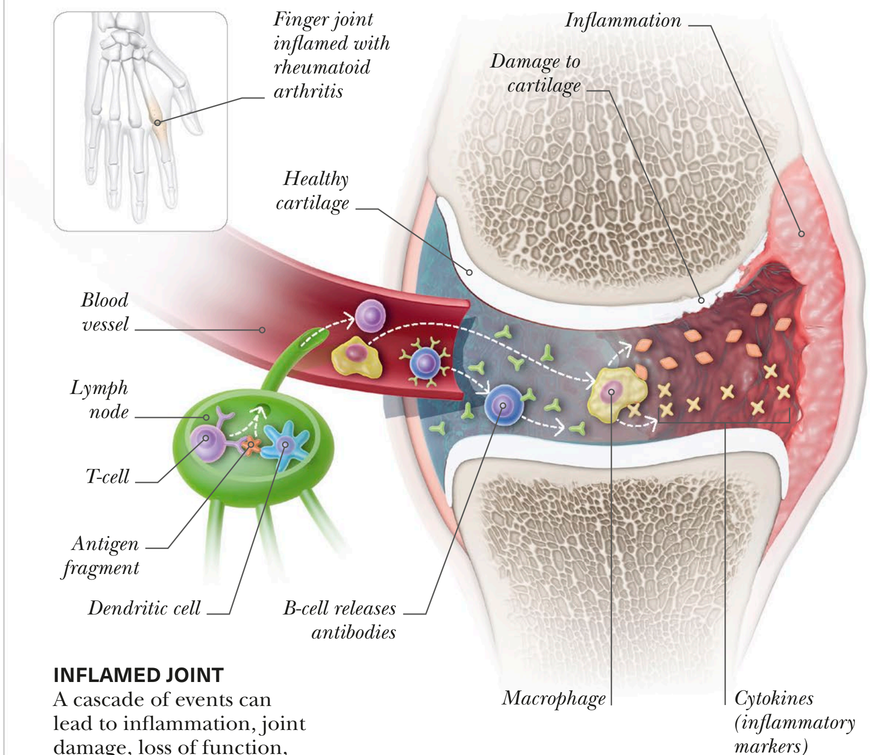
B-CELL
A type of lymphocyte that secretes antibodies, which are proteins specialized to fight specific antigens.

T-CELL
A type of lymphocyte that is activated to fight by the presentation of antigens. There are many specialized types.

PHAGOCYTOSIS
Macrophages (white) patrol your body on alert for invaders (red) to engulf and eat, in a process called phagocytosis.

INFLAMMATORY RESPONSE

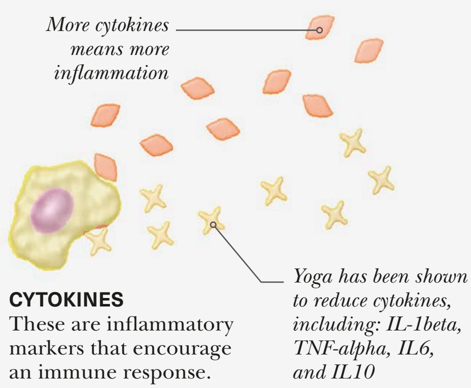
Inflammation often involves heat, pain, redness, and swelling due to a cascade of events where white blood cells fight invaders. In an autoimmune disease, they mistakenly fight body tissue. For example, rheumatoid arthritis (see below) can flare to cause local inflammation and body-wide inflammation.



INFLAMED JOINT
A cascade of events can lead to inflammation, joint damage, loss of function, and possibly pain.

Yoga and inflammation

Yoga seems to help attenuate inflammation by reducing the stress response, which may reduce your disease risk. A review shows that yoga practice reduces cytokine count and therefore inflammation. Scientists hypothesize that a long-term, regular practice would be most effective.



CYTOKINES
These are inflammatory markers that encourage an immune response.